

Question	Answer	Marks
9(a)(i)	Hall voltage depends on thickness of slice	C1
	thinner slice, larger Hall voltage	A1
9(a)(ii)	Hall voltage depends on current in slice	B1
9(b)	sinusoidal wave, one cycle	B1
	at $\theta = 0$ and at $\theta = 360^\circ$, $V_H = V_{MAX}$	B1
	at $\theta = 180^\circ$, $V_H = -V_{MAX}$	B1

Question	Answer	Marks
10(a)	two from: • frequency below which electrons not ejected • <u>maximum</u> energy of electron depends on frequency • <u>maximum</u> energy of electrons does not depend on intensity • instantaneous emission of electrons	B2
10(b)(i)	(λ_0 is the) threshold wavelength or wavelength corresponding to threshold frequency or maximum wavelength for emission of electrons	B1
10(b)(ii)1.	intercept = $1/\lambda_0 = 2.2 \times 10^6 \text{ m}^{-1}$ $\lambda_0 = 4.5 \times 10^{-7} \text{ m}$ or 450 nm	A1
10(b)(ii)2.	gradient = hc	C1
	gradient = 2.0×10^{-25} or correct substitution into gradient formula	C1
	$h = (2.0 \times 10^{-25}) / (3.0 \times 10^8) = 6.7 \times 10^{-34} \text{ J s}$	A1
10(c)	line: same gradient	B1
	straight line, positive gradient, intercept at greater than 2.2×10^6 when candidate's line extrapolated	B1